Nonstandard Analysis and Certified Presentation Trading as Sources of Proof-Search Speedups

From Unlimited Derivability to Budget-Limited Settlement in ATPs, SICs, and AMLDs

Brian Tenneson
Research synthesis and theorem-development memorandum

September 2, 2026

Abstract

Classical provability asks whether a certificate exists; an actual automated theorem prover (ATP), settlement-in-context system (SIC), or agentic mathematical logic discovery system (AMLD) asks whether a certificate can be found and verified inside a finite budget. This paper develops nonstandard analysis (NSA) and certified presentation trading as a bridge between those questions. Trading makes the presentation itself a search variable: one may exchange inference rules, axioms, definitions, macros, quotients, or lemmas while preserving the relevant consequence closure and retaining a certified return map to the protected original presentation. Transfer, overspill, underspill, saturation, standard part, Loeb measure, nonstandard hulls, proof mining, and nonstandard models of arithmetic then operate over a joint space of presentations and proof strategies. We assemble classical speed-up and proof-complexity results, distinguish semantic equivalence from budgeted nonequivalence, and formulate verifier-safe conditional speed-up theorems. The principal message is: NSA-guided trading becomes operational only after extraction. A hyperfinite witness or advantageous traded presentation is semantic compression; a speed-up theorem requires a standard extractor, a complete cost model, and a certificate-preserving return translation.

Central thesis. Passing from unlimited to limited proof length reveals distinctions invisible to consequence closure. Certified trading preserves what is provable while changing how costly it is to discover and express a proof. NSA supplies existence, quotient, stabilization, measure, and extraction principles for searching the resulting presentation–strategy space. When a selected trade has a standard computable return map to an independently checkable certificate, the combination can yield genuine instance-family speedups without changing the protected notion of correctness.

Contents

1 The problem: provability is not settlement

Let a presentation be $P = (\Sigma, A, R)$ with verifier V_P. Write

$$P \vdash \varphi \quad \Longleftrightarrow \quad \exists \pi \; V_P(\pi, \varphi) = 1.$$

This is unlimited derivability. For a resource vector $b = (t, m, e, \ell, q)$ (time, memory, expansions, certificate length, and query budget), define

$$\mathrm{Settle}_{P,\mathcal{A}}(\varphi; b) \in \{P, R, I, C, U\},$$

where $\mathcal{A}$ is the search architecture and U means unsettled within budget. The positive proof-length profile is

$$L_P(\varphi) = \min\{\mathrm{len}(\pi) : V_P(\pi, \varphi) = 1\},$$

with $L_P(\varphi) = \infty$ if no proof exists. Search cost and proof length are different:

$$L_P(\varphi) \leq b_\ell \not\Rightarrow \mathcal{A} \text{ finds such a proof within } b_t, b_m, b_e.$$

For AMLD, the certificate may settle proof, refutation, independence, or inconsistency. Accordingly define a settlement profile

$$\mathbf{L}_P(\varphi) = (L_P^+, L_P^-, L_P^I, L_P^C),$$

and compare architectures by the budgeted preorder

$$(P, \mathcal{A}) \preceq_{\mathcal{D},B} (Q, \mathcal{B}) \iff \Pr_{\varphi\sim\mathcal{D}}[\mathrm{Settle}_{P,\mathcal{A}}(\varphi; B) \neq U] \leq \Pr_{\varphi\sim\mathcal{D}}[\mathrm{Settle}_{Q,\mathcal{B}}(\varphi; B) \neq U].$$

This is the natural home for empirical speedup claims. It preserves the distinction between semantic equivalence, certificate size, and discovery cost.

Trading enlarges the search object. Let $\mathfrak{T}(P)$ be a family of presentations Q equipped with target translations $\sigma_{P\to Q}$ and certificate-return maps $\tau_{Q\to P}$. A certified trade requires

$$V_Q(\rho, \sigma_{P\to Q}(\varphi)) = 1 \quad \Longrightarrow \quad V_P(\tau_{Q\to P}(\rho), \varphi) = 1.$$

The protected trading architecture searches pairs

$$(Q, \mathcal{A}_Q) \in \mathfrak{T}(P) \times \mathfrak{A},$$

while retaining $(P, \mathcal{A}_P)$ as a fallback. Unlimited equivalence may give $\mathrm{Cn}(P) = \mathrm{Cn}(Q)$, yet limited settlement can differ. Thus trading preserves unlimited settlement while potentially changing limited settlement.

2 The classical speed-up landscape

NSA enters an area with strong existing positive and negative results. Any proposed NSA theorem must be placed against these baselines.

2.1 Unbounded recursive speedup

Gödel announced, and Buss later proved in the relevant formulations, that moving up an order of arithmetic can yield speedup exceeding any prescribed recursive function: for every recursive f, infinitely many common theorems have shortest lower-order proofs longer than f applied to their higher-order proof lengths [1, 2]. This is the canonical example of a stronger language compressing a family of proofs by expressing uniform truth at a higher type.

Ehrenfeucht and Mycielski proved a complementary axiom-extension phenomenon: under an undecidability hypothesis, adding an axiom can abbreviate proofs by more than any recursive bound [3]. These results directly support presentation trading in one direction: logically related presentations can have radically different finite-budget behavior.

2.2 Cuts, lemmas, and macros

Cut elimination preserves provability, but Orevkov–Statman families show that eliminating cuts can require nonelementary growth [4]. A cut or reusable lemma is therefore not mere convenience; it can be an enormous proof-compression device. In ATP language, a verified lemma bank is a controlled cut-introduction mechanism. In SIC/AMLD language, a BANK deposit can change future settlement cost without changing consequence closure.

Definitions and extension variables similarly abbreviate repeated structure. Extended Frege systems formalize this idea propositionally. The open status of strong lower bounds for Frege and extended Frege is a warning: one must not claim a general polynomial speedup from macro introduction without restrictions.

2.3 Cook–Reckhow proof systems and barriers

A Cook–Reckhow proof system is a polynomial-time computable surjection onto tautologies. A system is polynomially bounded when every tautology has a polynomial-size proof. Such a system exists iff NP = coNP [5]. Thus a universal guarantee that every theorem acquires a short, efficiently verifiable propositional certificate would resolve a central open problem. A proposed NSA optimizer can legitimately improve distributions or families, but a general polynomial boundedness claim carries this barrier.

Proof systems are compared by simulation and p-simulation. Whether an optimal or p-optimal propositional proof system exists remains open. A multi-presentation AMLD can approximate a portfolio optimum empirically; calling it universally optimal would be mathematically much stronger.

2.4 Finite consistency and bounded arithmetic

Finite consistency statements $\mathrm{Con}(T) \restriction n$ assert that no contradiction has a T-proof of size at most n. They make the unlimited-to-limited transition an object-level arithmetic statement. Work of Pudlák and others exhibits short proofs for some finite consistency families, while related strengthened consistency families are conjectured not to have polynomial proofs [8, 9]. Parikh's feasibility program and bounded arithmetic isolate which existence proofs supply feasible bounds [6, 7].

This yields a key design lesson: the bounded statement

$$\neg\exists\pi\,[\mathrm{len}(\pi) \leq n \wedge V_T(\pi, \bot) = 1]$$

is not merely a truncated version of $\mathrm{Con}(T)$; its proof complexity is a diagnostic of the settlement architecture itself.

3 Certified presentation trading

3.1 Rules, axioms, definitions, macros, and quotients

Write a formal system as $T_1(x, y, z)$, where x is the language, y the axioms, and z a set of inference rules. A rule-to-axiom trading map t seeks a presentation

$$T_2(x, y \cup t(z), \varnothing)$$

such that

$$\mathrm{Cn}(T_1(x,y,z)) = \mathrm{Cn}(T_2(x, y \cup t(z), \varnothing)).$$

The rules on the left are represented axiomatically on the right; conversely, axioms may be packaged as admissible rules or macros. More general trades can introduce definitions, quotient objects, extension variables, verified lemmas, or alternative encodings. The equality of consequence closures is semantic. It says nothing by itself about proof length, branching, unification, indexing, learning features, or translation cost.

Definition 3.1 (Certified trade)**.** *A certified trade from P to Q on a target class Φ consists of (σ, τ, e) such that σ translates targets, τ translates successful Q-certificates back to P, and e certifies for every $\varphi \in \Phi$ that*

$$V_Q(\rho, \sigma(\varphi)) = 1 \Longrightarrow V_P(\tau(\rho), \varphi) = 1.$$

Full equivalence is desirable but not necessary when a certified one-way return map suffices for settlement.

3.2 The complete cost of a trade

For a Q-certificate ρ, define

$$C_{P \rightsquigarrow Q}(\varphi, \rho) = C_{\text{select}}(P, Q, \varphi) + C_{\text{search}}(Q, \sigma(\varphi)) + C_{\text{return}}(\rho) + C_{\text{verify}}(\tau(\rho)).$$

The traded proof-length analogue is

$$L_{P \rightsquigarrow Q}(\varphi) = c(P,Q) + L_Q(\sigma(\varphi)) + \mathrm{len}(\tau(\rho_Q)),$$

where ρ_Q is a shortest or selected Q-certificate and $c(P,Q)$ accounts for representing and certifying the trade. The protected traded optimum is

$$L_{\mathfrak{T}(P)}(\varphi) = \min \left\{ L_P(\varphi), \inf_{Q \in \mathfrak{T}(P)} L_{P \rightsquigarrow Q}(\varphi) \right\}.$$

Including L_P formalizes baseline protection: the architecture may add trades but need not withdraw the original presentation.

3.3 Trading configurations and the BANK

Let Z be a finite or hyperfinite set of tradable rules, axioms, definitions, lemmas, or quotient relations. A configuration may be encoded by $\theta \in \{0,1,2\}^Z$, with 0 meaning original form, 1 retaining both forms, and 2 prioritizing the traded form while preserving a fallback. A multi-resource objective is

$$F_\varphi(\theta) = \alpha T_\varphi(\theta) + \beta M_\varphi(\theta) + \gamma E_\varphi(\theta) + \eta L_\varphi(\theta) + \kappa C_{\text{return}}(\theta).$$

The BANK may store verified trading instruments

$$\mathcal{B}_T = \{(P, Q, \sigma, \tau, e, \mathbf{c}, \mathcal{D}_{\text{evidence}})\},$$

including provenance, observed target families, resource vectors, benefits, regressions, and the return certificate. Such entries are deposit-only knowledge; using a trade never deletes the original rules or axioms.

4 NSA principles relevant to speedup

Fix a sufficiently saturated enlargement $*\mathcal{U}$ and distinguish internal from external objects.

4.1 Transfer: one verified algorithm, standard and hyperfinite

Transfer sends first-order statements true in the standard structure to their star-transforms. A bounded search predicate becomes

$$(\exists n \leq N)\, V^*(\pi_n, \varphi) = 1$$

for hypernatural N. Transfer is valuable for correctness proofs: the same invariant can govern finite and hyperfinite executions. But transfer alone gives no standard running-time improvement.

4.2 Overspill and underspill: extracting thresholds

Overspill says that an internal property holding for every standard natural holds up to some infinite hypernatural. Underspill gives a finite threshold from an internal property holding throughout an infinite tail (under the usual hypotheses). Applied to search:

- overspill turns all-standard-stage success into an internal long-horizon invariant;
- underspill turns uniform behavior beyond every infinite budget into the existence of a standard cutoff;
- neither supplies a computable cutoff unless the proof admits effective extraction.

4.3 Saturation: compatible local hints become a global object

Saturation realizes a type when its finite subfamilies are satisfiable. For an AMLD, local constraints supplied by agents can be viewed as a family $\{C_i : i \in I\}$. Saturation suggests a global latent strategy satisfying every standard finite subfamily. This is a principled semantic model for combining partial credit, quotient invariants, countermodels, and proof fragments. Operationally, however, the realized object may be external to any feasible computation.

4.4 Standard part and nonstandard hulls: quotienting noise

Finite hyperreals x, y with $x \approx y$ share a standard part. This induces a quotient by infinitesimal variation. For proof search, let $F(s)$ be a bounded feature vector or normalized cost. States with $F(s) \approx F(t)$ can be merged at the macroscopic level. A nonstandard hull similarly quotients finite elements by infinitesimal distance. This motivates approximate bisimulation and state aggregation, but correctness requires a lifting theorem from the quotient path to a verifier-accepted original certificate.

4.5 Loeb measure: probability on a hyperfinite portfolio

An internal counting measure on a hyperfinite set extends to a standard countably additive Loeb measure. This allows rigorous probability statements about a hyperfinite portfolio of strategies, traces, or presentations. It can turn empirical density into a standard probability and support almost-sure claims. It does not identify a fast individual strategy without selection or derandomization.

With trading, the natural portfolio is

$$H_P = \{(Q_i, \mathcal{A}_i, \tau_i) : i \leq N\}, \qquad N \in {}^*\mathbb{N},$$

where every internal member carries a return translator. For a standard target and budget, the successful region consists only of pairs whose returned certificate verifies in P. A positive Loeb success measure can justify sampling or compression of the portfolio, but a standard extractor must still produce a finite computable support.

4.6 Internal definition, idealization, and standardization

Nelson-style Internal Set Theory packages NSA through transfer, idealization, and standardization. Idealization resembles the passage from every standard finite collection of requirements to one object meeting them all; standardization extracts a standard trace of a possibly external class. For ATP use, the important question is conservativity: if the NSA formalism is conservative over the base theory for the target class, it may shorten reasoning without adding standard theorems. Conservativity alone does not bound the translation blowup.

4.7 Proof mining and computational content

Proof mining extracts rates, moduli, witnesses, and algorithms from proofs. Work on the computational content of NSA shows that substantial classes of nonstandard arguments admit term extraction, including "pure" nonstandard formulations [14]. The recurring normal form is

$$(\forall^{\mathrm{st}} x)(\exists^{\mathrm{st}} y)\, \psi(x, y), \qquad \psi \text{ internal},$$

from which one seeks a standard functional t such that

$$(\forall x)(\exists y \in t(x))\, \psi(x, y).$$

For automated settlement, $t(x)$ is a finite candidate list, strategy portfolio, cutoff, lemma set, or quotient cover. This is the most direct rigorous route from NSA existence to an executable speedup.

4.8 Metastability instead of convergence

The metastable form of convergence replaces an asymptotic statement by

$$\forall k\, \forall g\, \exists N\, \forall i, j \in [N, N + g(N)] \quad d(x_i, x_j) < 2^{-k}.$$

This is already budget-shaped: it seeks a finite window of stability. For learned schedulers and iterative strategy weights, a metastability bound can become a stopping rule, avoiding an uncomputable or excessively conservative demand for full convergence.

In a trading system, x_n may encode rankings of presentation–strategy pairs rather than strategies alone. Metastability then certifies a finite window during which the system can stop revising which rules, axioms, quotients, or macros it prioritizes.

5 Verifier-safe conditional speed-up theorems

The following propositions isolate the hypotheses needed to convert NSA intuition into actual bounded settlement gains.

Theorem 5.1 (Extraction-to-settlement)**.** *Suppose an NSA proof establishes*

$$(\forall^{\mathrm{st}} x)(\exists^{\mathrm{st}} y)\,[R(x,y) \wedge \mathrm{Cost}(y) \le q(|x|)],$$

where R is internal and decidable by a verifier in time $v(|x|+|y|)$. If term extraction yields a standard finite list $t(x)$ containing a suitable y, computable in time $e(|x|)$, then exhaustive verification of $t(x)$ settles every standard instance in time

$$e(|x|) + |t(x)|\, v(|x| + q(|x|)).$$

Relative to any baseline with lower bound $B(|x|)$, this is a speedup wherever the displayed bound is $o(B(|x|))$.

Proof. Compute the extracted list and verify each member. Soundness is inherited from the verifier, not from trust in the nonstandard argument. The cost bound is immediate. □

Theorem 5.2 (Hyperfinite portfolio compression)**.** *Let H be a hyperfinite internal family of strategies with Loeb measure μ_L. Suppose an internal success set $G_x \subseteq H$ satisfies $\mu_L(G_x) \ge \delta > 0$ for every standard x, and suppose a standard sampler approximates the Loeb distribution within total variation $\epsilon < \delta$. Then k independent sampled strategies miss G_x with probability at most $(1-\delta+\epsilon)^k$.*

This is a genuine probabilistic amplification theorem, not yet a proof-length theorem. It becomes one when each successful run returns a checkable certificate and the per-run budget is bounded.

Theorem 5.3 (Quotient-lifting speedup)**.** *Let $q : S \to \bar{S}$ map proof-search states to quotient states. Assume: (i) every quotient transition has a standard computable lift of cost at most $a(n)$; (ii) every quotient certificate of length r lifts to an original certificate of length at most $a(n)r + b(n)$; and (iii) quotient search visits at most $M(n)$ states versus a baseline lower bound $N(n)$. Then quotient-first settlement has discovery cost $O(M(n) + a(n)r + b(n))$ and is asymptotically faster whenever this is $o(N(n))$.*

NSA's standard-part quotient motivates q; hypotheses (i)–(ii) are the essential verifier-safety obligations.

Theorem 5.4 (Metastable scheduler stopping)**.** *Let w_n be the strategy-weight vector and suppose proof mining extracts $\Phi(k,g)$ satisfying metastability. If the action-ranking map is margin-stable whenever $\|w_i - w_j\| < 2^{-k}$, then after searching $N \le \Phi(k,g)$ the scheduler can freeze its ranking for the next $g(N)$ updates without changing selected actions.*

This converts an asymptotic learning claim into a finite operational pause rule.

Proposition 5.5 (BANK as cut compression)**.** *If a verified bank lemma λ has acquisition cost $c(\lambda)$ and replaces m repeated subderivations of lengths $d_1, \ldots, d_m$ by references of cost r_i, then its net certificate saving is*

$$\Delta(\lambda) = \sum_{i=1}^{m} (d_i - r_i) - c(\lambda).$$

Deposit is proof-length beneficial exactly when $\Delta(\lambda) > 0$. Under a hyperfinite workload, Loeb integration of Δ defines an expected standard utility whenever the normalized utility is Loeb integrable.

Theorem 5.6 (Extracted certified trading speedup)**.** *Let P be a standard protected presentation and $\mathfrak{T}(P)$ an internally represented family of certified trades. Suppose an NSA proof establishes that for every standard target x there are standard $Q \in \mathfrak{T}(P)$ and Q-certificate ρ whose complete returned settlement cost is at most $q(|x|)$. If functional interpretation extracts a finite standard list*

$$\mathcal{L}(x) = \{(Q_1, \rho_1, \tau_1), \ldots, (Q_k, \rho_k, \tau_k)\}$$

containing a successful member, and every τ_i is computable and verifier-sound, then checking the list is a standard P-settler. Its cost is the extraction cost plus the sum of trade checking, Q_i verification, return translation, and P verification costs. It is an asymptotic speedup over direct P settlement on a family x_n whenever this total is $o(B_P(x_n))$ for a valid baseline lower bound B_P.

Proof. Compute the extracted list. For each entry verify the trade certificate and the Q_i-certificate, translate it, and accept only if the protected verifier V_P accepts. Existence of a successful list member gives completeness on the stated target family; V_P gives soundness. The cost and speedup conclusions follow by addition and comparison. □

Proposition 5.7 (Translation inequality)**.** *For one certified trade $P \rightsquigarrow Q$, if*

$$\mathrm{len}(\tau(\pi_Q)) \leq a(\mathrm{len}(\pi_Q)) + c(|\varphi|),$$

then Q-search followed by translation yields an instance-family proof-length speedup whenever

$$a(L_Q(\sigma(\varphi_n))) + c(|\varphi_n|) = o(L_P(\varphi_n)).$$

NSA may help select Q by quotient, saturation, or hyperfinite portfolio methods; the speedup rests on this standard inequality.

6 How each NSA tool maps into ATP/SIC/AMLD architecture

NSA device	Mathematical role	Operational counterpart / obligation
Transfer	move finite laws to internal hyperfinite domains	reuse invariants; no speed claim without extraction
Overspill	extend all-standard behavior	detect persistent regime; cutoff may be noncomputable
Underspill	obtain a standard threshold	budget certificate if an effective bound is extracted
Saturation	realize finitely compatible constraints	combine agents/hints; requires finite representation
Standard part	quotient infinitesimal differences	state merging; must lift certificates
Nonstandard hull	complete/quotient a metric space	cluster proof states or presentations
Loeb measure	standard probability from hyperfinite counting	portfolio allocation and settlement density
Idealization	exchange finite-family and uniform witnesses	multi-agent constraint aggregation
Proof mining	extract finite lists/rates/moduli	executable search plan or stopping bound
Metastability	finitary surrogate for convergence	freeze learner/scheduler over certified window
Nonstandard arithmetic	encode long finite behavior internally	finite consistency and lower-bound diagnostics

7 A proposed NSA speedup program

7.1 The NSA–trading settlement compiler

For a target family $\mathcal{F}$, use five stages:

1. **Internalize.** Express proof search, presentation trades, return translation, resource counters, and BANK operations by internal formulas.
2. **Compress semantically.** Apply saturation, standard part, a hull, or Loeb measure to prove existence of a good presentation–strategy pair or portfolio.
3. **Normalize.** Rewrite the result into a standard-existential normal form suitable for term extraction.
4. **Extract.** Produce a finite list of certified trades, rates, cutoffs, quotient covers, or schedulers.
5. **Return, certify, and benchmark.** Translate every success to the protected original presentation, verify there, and compare against the retained baseline under identical complete budgets.

7.2 A falsifiable principal conjecture

Conjecture 7.1 (Certified NSA–trading speedup)**.** *There exist a natural held-out distribution $\mathcal{D}$, a protected presentation P, and an NSA-derived, standard-extracted finite trading policy Π_{NSA} such that, with equal complete resource budget B,*

$$\Pr_{\varphi\sim\mathcal{D}}[\mathrm{Settle}^{\mathrm{trade}}_{P,\Pi_{NSA}}(\varphi;B)\neq U] > \Pr_{\varphi\sim\mathcal{D}}[\mathrm{Settle}_{P,\mathcal{A}_0}(\varphi;B)\neq U],$$

with a preregistered positive margin, while every selected trade is certified on the target class, every returned certificate verifies in P, selection and return costs are charged, P remains available as fallback, and the gain persists on targets not used to select trades.

This is deliberately weaker than a universal proof-complexity claim and stronger than an anecdotal solve.

7.3 Experiments that discriminate mechanisms

1. **Metastability ablation:** extracted freeze rule versus fixed patience and convergence heuristics.
2. **Quotient lifting:** raw state search versus standard-part-inspired clustering, measuring lost paths, lift cost, and verified solves.
3. **Loeb trading-portfolio surrogate:** increasing finite portfolios of presentation–strategy pairs approximating an internal distribution; test whether settlement density stabilizes and whether a small extracted support preserves it after return costs.
4. **Saturation-to-list extraction:** local agent constraints converted into a finite candidate lemma list; compare list size and hit rate.
5. **Finite consistency diagnostics:** generate $\mathrm{Con}(T) \restriction n$ families and measure sensitivity to presentation trading and BANK cuts.
6. **Rule/axiom trading:** compare original rules, traded axioms, and the retained-both configuration; report rule matching, indexing, proof length, and expansion back to the original.
7. **Protected trading:** search certified equivalent presentations but always retain the original. Charge selection, trade verification, discovery, return translation, and final verification separately.

8 What would count as a theorem, and what would not

Warning 8.1. *An infinite hypernatural budget is larger than every standard finite budget. Solving by stage $H \in {}^*\mathbb{N}\setminus\mathbb{N}$ does not imply a feasible standard algorithm.*

Warning 8.2. *Compactness and saturation often prove existence nonconstructively. They do not automatically produce a certificate, strategy, or computable bound.*

Warning 8.3. *A shorter proof is not necessarily easier to find; a fast search can return a long certificate; a compressed certificate may be expensive to decompress. Report all three costs.*

Warning 8.4. *Standard-part maps are generally external. Treating* st *as a freely computable instruction is invalid.*

Warning 8.5. *Distributional gains do not establish worst-case proof-system simulation. Conversely, worst-case barriers do not prohibit large practical gains on structured distributions.*

9 Synthesis

The most promising interactions are not confined to hyperfinite enumeration. Trading supplies the presentation space on which NSA can perform selection and compression.

1. **Proof mining is the execution bridge**: NSA proves a uniform standard witness exists; extraction turns it into a finite portfolio of trades and strategies.
2. **Standard part is the quotient bridge**: infinitesimally equivalent states or presentations can be collapsed, provided certificates lift and return.
3. **Loeb measure is the portfolio bridge**: hyperfinite presentation–strategy populations become standard probability distributions and small-support targets.
4. **Metastability is the stopping bridge**: asymptotic learning is replaced with a finite window in which trade rankings stabilize.
5. **Nonstandard arithmetic is the diagnostic bridge**: bounded proof and finite-consistency statements expose where a presentation or theory hides cost.
6. **Classical speedup theory is the guardrail**: arbitrary recursive speedups are possible between theories/presentations, yet universal efficient certificates face Cook–Reckhow barriers.

For ATPs, the near-term target is extracted candidate and presentation reduction. For SICs, it is contextual trading and quotienting with a lift. For AMLDs, it is hyperfinite/Loeb reasoning over presentation–strategy pairs, metastable scheduling, and a BANK of verified trading instruments. Across all three, the invariant is the same: NSA and trading may guide discovery, but the protected standard verifier settles the claim.

10 Conclusion

The unlimited statement $P \vdash \varphi$ erases the very phenomenon an actual prover must manage. Trading makes this visible: presentations with the same consequence closure can have radically different budgeted settlement profiles. The correct object is therefore a budget-indexed profile over a protected, certified trading space. NSA offers semantic compression devices—transfer, spill principles, saturation, quotients, Loeb probability, idealization, extraction, and metastability—for navigating that space. Their use is mathematically serious only when explicit standard extraction and return translation close the loop.

The strongest defensible thesis is therefore conditional but testable: *NSA-derived semantic compression can select advantageous certified presentations and produce finite-budget settlement speedups when the selection is effectively extracted, all trading costs are charged, and every compressed path returns to an independently verified certificate in the protected original presentation.* This encompasses, but is much broader than, hyperfinite search.

References

[1] K. Gödel, "Über die Länge von Beweisen," Ergebnisse eines mathematischen Kolloquiums 7 (1936), 23–24.

[2] S. R. Buss, "On Gödel's theorems on lengths of proofs I: Number of lines and speedup for arithmetics," J. Symbolic Logic 59 (1994), 737–756.

[3] A. Ehrenfeucht and J. Mycielski, "Abbreviating proofs by adding new axioms," Bull. Amer. Math. Soc. 77 (1971), 366–367.

[4] M. Baaz and A. Leitsch, "Cut normal forms and proof complexity," Ann. Pure Appl. Logic 97 (1999), 127–177.

[5] S. A. Cook and R. A. Reckhow, "The relative efficiency of propositional proof systems," J. Symbolic Logic 44 (1979), 36–50.

[6] R. Parikh, "Existence and feasibility in arithmetic," J. Symbolic Logic 36 (1971), 494–508; and "Some results on the length of proofs," Trans. Amer. Math. Soc. 177 (1973), 29–36.

[7] S. R. Buss, "Bounded arithmetic, proof complexity and two papers of Parikh," Ann. Pure Appl. Logic 96 (1999), 43–55.

[8] P. Pudlák, "On the length of proofs of finitistic consistency statements in first order theories," in Logic Colloquium '84, North-Holland (1986), 165–196.

[9] A. Freund and F. Pakhomov, "Short proofs for slow consistency," Proc. Amer. Math. Soc. 148 (2020), 5377–5390.

[10] A. Robinson, *Non-standard Analysis*, North-Holland, 1966.

[11] E. Nelson, "Internal set theory: a new approach to nonstandard analysis," Bull. Amer. Math. Soc. 83 (1977), 1165–1198.

[12] P. A. Loeb, "Conversion from nonstandard to standard measure spaces and applications in probability theory," Trans. Amer. Math. Soc. 211 (1975), 113–122.

[13] U. Kohlenbach, *Applied Proof Theory: Proof Interpretations and Their Use in Mathematics*, Springer, 2008.

[14] S. Sanders, "The computational content of nonstandard analysis," in *Nonstandard Analysis for the Working Mathematician*, 2015/2016 preprint, arXiv:1606.06386.

[15] T. Tao, "Norm convergence of multiple ergodic averages for commuting transformations," Ergodic Theory Dynam. Systems 28 (2008), 657–688 (metastability viewpoint discussed in proof-mining literature).

[16] J. Krajíček, *Bounded Arithmetic, Propositional Logic, and Complexity Theory*, Cambridge Univ. Press, 1995.